Enzyme-discovery agent — retrieval-expansion architecture

Target chemistry drives discovery; the LLM proposes strategies, tools find entities (data-leakage firewall). Numbers from run 20260914_220431.

1 Target reaction contract

13 claims · 9 inferred

Derive chemistry from the query: direction, bond, reactive groups, activation options

2 Literature harvest

4 seed candidates

Read primary full text; seed DB expansion with tested enzymes before search

3 Mechanism-family hypotheses

6 families (3 high-promise)

Propose falsifiable mechanism folds to cover (with disqualifying observations)

4 Family survey + expansion

51 families · 1,858 discovered

Map InterPro folds; promote substrate-fit members; literature-guided expansion

5 Candidate evaluation

85 evaluated · 77 rejected

Per-candidate: EC/bond gate, substrate-fit axes, mechanism, cofactor spec, provenance

6 Decomposed ranking

38 ranked · 3 strong leads

Lexicographic spine over substrates_hosted / shape_mismatches / weakest_fg_match

7 Structural-modeling handoff

20 candidates

Per-candidate: family + mechanism (reactive P) + cofactor spec → co-fold

cross-cutting design

Data-leakage firewall

Answer enzymes never enter prompts/tools; only generic mechanism vocabulary is allowed.

Deterministic chemistry

Activation mode & cofactors derived from reaction products — no LLM, fully provenanced.

Decomposed, no composites

Substrate fit reasoned as functional-group facts, not fused Tanimoto scores.

Structural handoff

Explicit, reviewable spec: family + mechanism + cofactors travel to co-folding.